

Autonomic Nervous System Pharmacology

1-Introduction

Sympathetic and parasympathetic pathways

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Learning outcomes

This lecture connects autonomic pathways with receptor responses, then applies those mechanisms to drug selection.

By the end, students can...

- Describe ANS divisions and their anatomy.
- Explain transmitters and receptor families.
- Predict responses from receptor and organ.
- Classify drugs and apply them clinically.



Session roadmap

Pathways, receptors and clinical application

- 1 ANS anatomy and transmitter pathways
- 2 Receptor families and organ responses
- 3 Autonomic reflexes and drug classes
- 4 Clinical application and take-home messages

The physiological scope

The ANS adjusts visceral function to maintain homeostasis

- Automatic CNS control of heart, vessels and bronchi.
- Also regulates gut, glands, pupil, temperature and bladder.
- Rapid visceral adjustments support homeostasis.

Somatic and autonomic motor output

An autonomic ganglion relays and distributes the signal

Somatic output reaches skeletal muscle through one neuron. Autonomic output includes a ganglion, where the signal can diverge before reaching a visceral organ.

Somatic

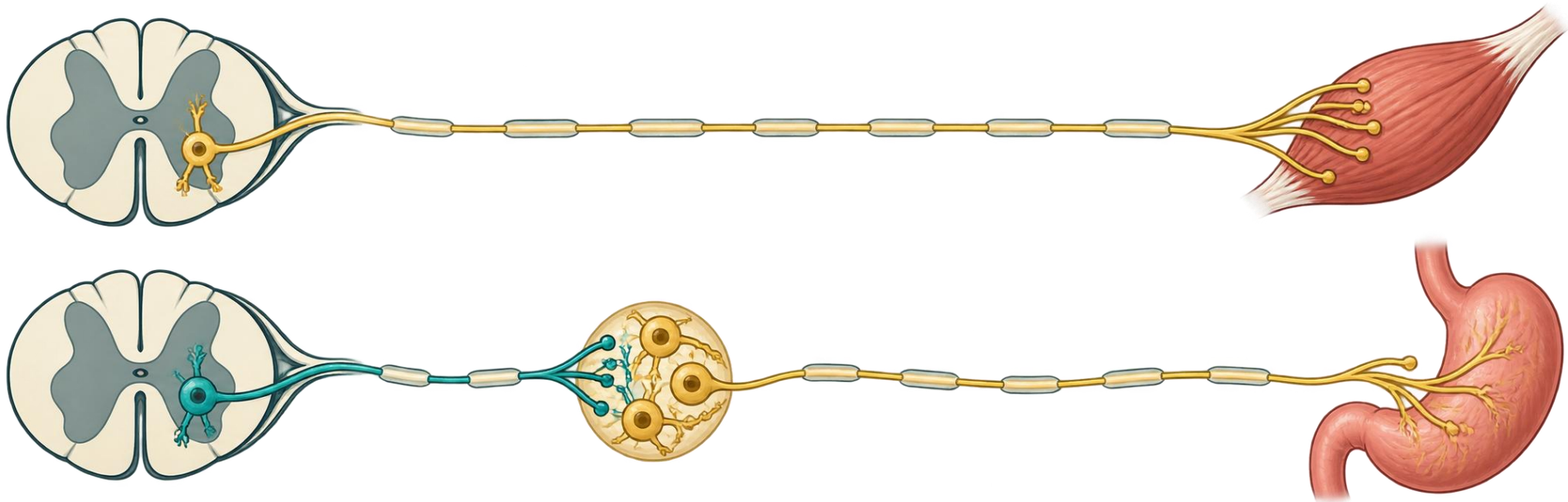
- One motor neuron runs from CNS to skeletal muscle.
- Nm receptors at the neuromuscular junction; usually voluntary.

Autonomic

- Preganglionic neuron → ganglion → postganglionic neuron → effector.
- Effector is cardiac muscle, smooth muscle or a gland.
- Ganglia permit amplification, divergence and local modulation.

Motor output: direct and ganglion-relayed paths

Why the autonomic signal passes through a ganglion



Top: one somatic motor neuron. Bottom: preganglionic neuron → ganglion → postganglionic neuron.

Sympathetic versus parasympathetic anatomy

Sympathetic • thoracolumbar

- T1–L2 spinal origin; paravertebral chain or prevertebral ganglia.
- Short preganglionic, long postganglionic fibers.
- Mobilizes energy and supports fight-or-flight responses.

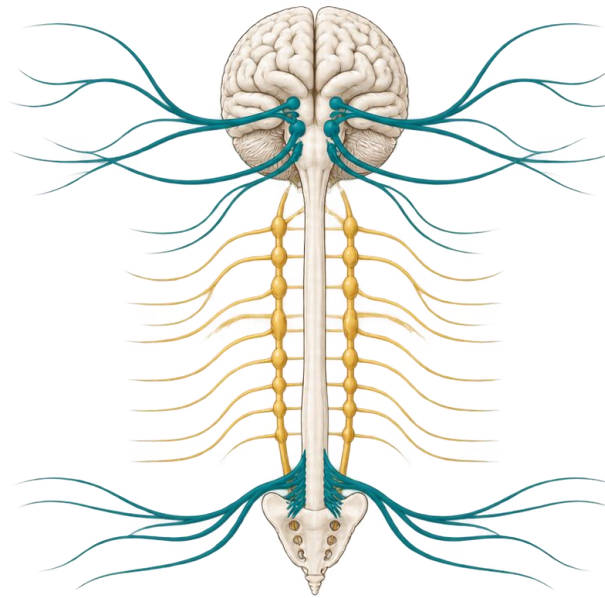
Parasympathetic • craniosacral

- CN III, VII, IX, X and S2–S4 outflow.
- Ganglia near or within target organ; long pre-, short postganglionic fibers.
- Supports digestion, secretion, micturition and reproductive functions.

Clinical link Failure of sympathetic vasoconstriction can contribute to orthostatic hypotension.

Origins of autonomic outflow

The origin helps predict the route to each ganglion



Sympathetic • T1–L2

Parasympathetic • CN III, VII, IX, X; S2–S4

Enteric nervous system

The enteric nervous system generates local gut reflexes.

Enteric nervous system

- Myenteric and submucosal plexuses lie in the gut wall.
- Local reflexes control motility, secretion and intestinal blood flow.
- Autonomic input modulates the circuits.

Adrenal medulla

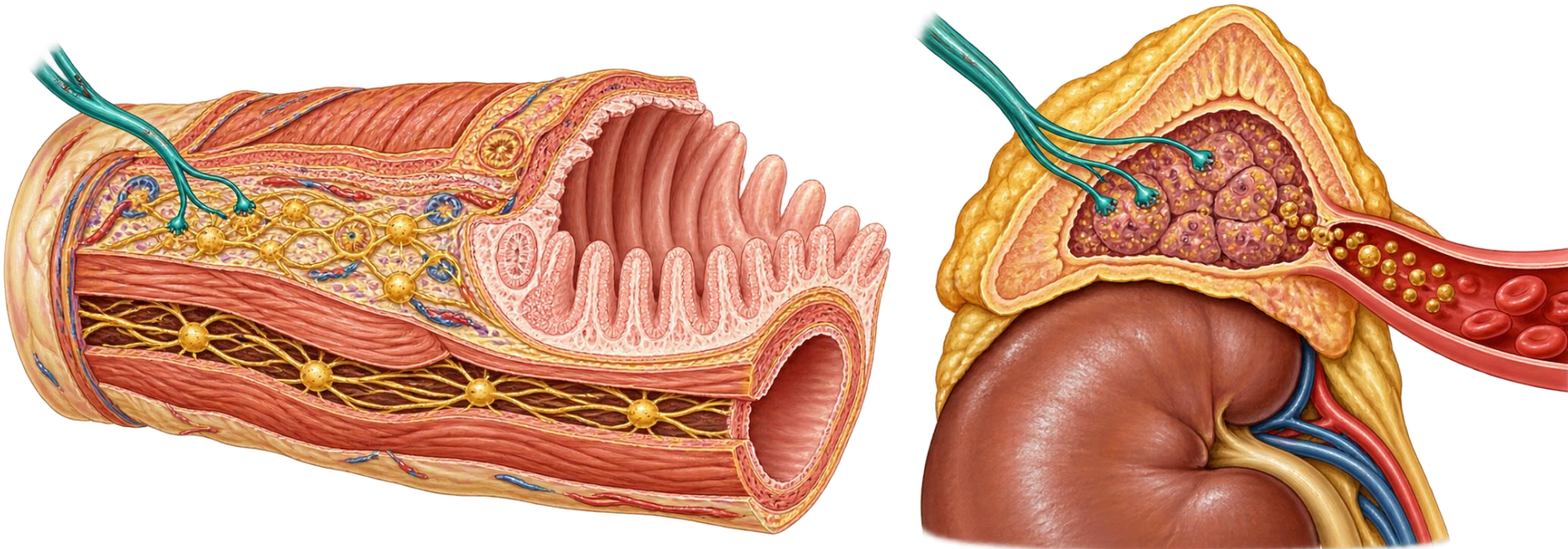
The adrenal medulla distributes a sympathetic signal through the bloodstream.

Adrenal medulla

- Preganglionic sympathetic ACh acts on chromaffin cells.
- It functions as a modified ganglion without a postganglionic axon.
- Mainly ***epinephrine*** and some ***norepinephrine*** enter the blood.

Enteric circuits and adrenal medulla

Local gut reflexes and a circulating sympathetic signal



Enteric: myenteric + submucosal plexuses in the gut wall; autonomic input modifies local reflexes.

Adrenal: preganglionic ACh → Nn chromaffin cells → circulating **epinephrine** and **norepinephrine**.



Transmitters along the two pathways

Both divisions begin with acetylcholine at Nn receptors. The transmitter released at the effector then determines the next receptor.

Sympathetic

- CNS → ACh → Nn ganglion → usually NE → α/β effector.
- Adrenal chromaffin cells release **epinephrine** and NE to blood.
- Other modulators: dopamine, VIP, nitric oxide and neuropeptide Y.

Parasympathetic

- CNS → ACh → Nn ganglion → ACh → muscarinic effector.
- Most postganglionic parasympathetic neurons release ACh.
- Transmitter identity and target receptor must be considered together.

Transmitter maps across motor pathways

Follow each signal from the CNS to its target

Pathway	Signal from CNS to target
Sympathetic	ACh → Nn ganglion → usually NE → adrenergic α or β receptor.
Parasympathetic	ACh → Nn ganglion → ACh → muscarinic receptor.
Adrenal medulla	ACh → Nn chromaffin cell → circulating epinephrine and NE.
Somatic	One motor neuron releases ACh onto Nm at skeletal muscle.



Exceptions to the usual sympathetic map

Anatomical origin does not always identify the effector transmitter

A sympathetic pathway does not always end with ***noradrenaline*** at the target. The transmitter and effector receptor must both be identified.

Site / concept	Important exceptions
Sweat glands	Sympathetic postganglionic ACh activates muscarinic receptors.
Adrenal medulla	Preganglionic ACh activates Nn on chromaffin cells; catecholamines enter blood.
Renal vascular bed	Dopamine receptors participate in renal vascular pharmacology.
Usual pattern	Most sympathetic effectors receive NE, but the exceptions matter clinically.



Receptor families and signaling

Fast ion channels and slower GPCR pathways

Nicotinic receptors are ion channels with a rapid response; muscarinic and adrenergic receptors use G-protein signaling.

Cholinergic

- Nicotinic Nn/Nm: ligand-gated ion channels with fast depolarization.
- Muscarinic M1–M5: GPCRs for ACh at many effector sites.

Adrenergic and other GPCRs

- α and β receptors respond to NE and ***epinephrine***.
- Dopaminergic receptors matter in vascular and CNS pharmacology.
- GPCR second messengers amplify responses over seconds to minutes.

Nicotinic receptors: Nn and Nm

The subtype indicates the site and function

Acetylcholine activates both nicotinic subtypes. The location of the receptor distinguishes ganglionic transmission from skeletal muscle contraction.

Nn • neuronal	Nm • muscle
Autonomic ganglia in both sympathetic and parasympathetic routes.	Somatic neuromuscular junction.
Adrenal medulla chromaffin cells.	ACh triggers skeletal muscle contraction.
ACh causes rapid ganglionic transmission.	Both Nn and Nm are ligand-gated ion channels.

Muscarinic receptor subtypes

M2 slows the heart; M3 contracts smooth muscle or stimulates glands

Muscarinic subtypes respond to acetylcholine, but tissue distribution determines the physiological effect.

Subtype	Subtype effects
M1	Neuronal and gastric sites; supports gastric acid secretion.
M2	Heart; slows heart rate and AV conduction. Atropine blocks this action.
M3	Gland secretion and smooth muscle contraction, including bronchi and detrusor.
Family	Muscarinic receptors are GPCRs; organ location defines the response.

Clinical link An antimuscarinic may treat urgency while causing dry mouth through M3 blockade.



Alpha adrenergic receptors

$\alpha 1$ acts mainly after release; $\alpha 2$ limits transmitter release

Alpha receptors serve different roles: $\alpha 1$ commonly acts at smooth muscle, while $\alpha 2$ often limits further transmitter release.

$\alpha 1$

- Vascular smooth muscle: vasoconstriction.
- Radial iris: mydriasis; bladder neck: outlet contraction.
- ***Phenylephrine*** activates $\alpha 1$; ***prazosin*** and ***tamsulosin*** block $\alpha 1$.
- $\alpha 1$ blockade can lower BP or relax the prostate and bladder outlet.

$\alpha 2$

- Presynaptic terminals: reduced NE release.
- Central sites: reduced sympathetic outflow.

Beta adrenergic receptors

The receptor number predicts the dominant target

Beta receptor subtype and tissue location explain why one agonist stimulates the heart while another relaxes bronchial smooth muscle.

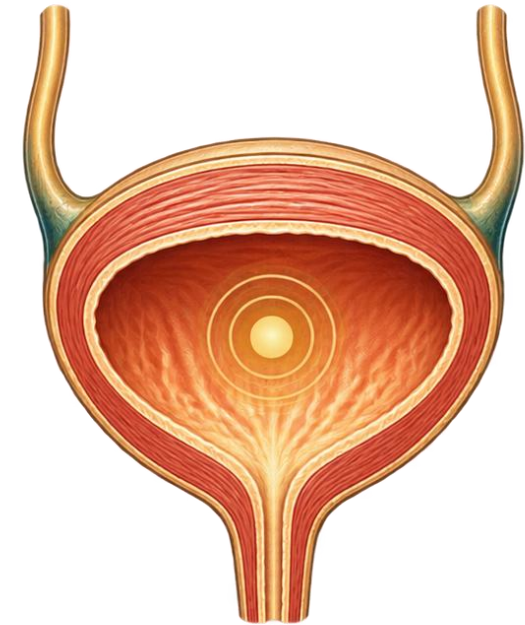
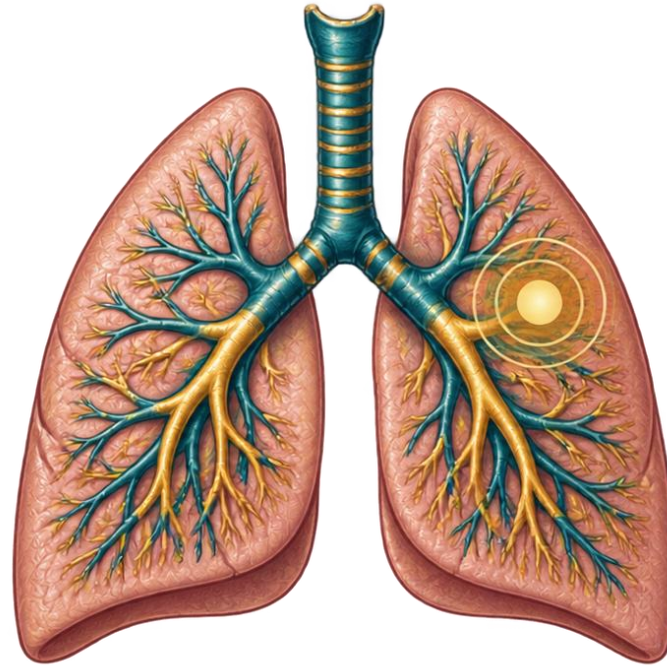
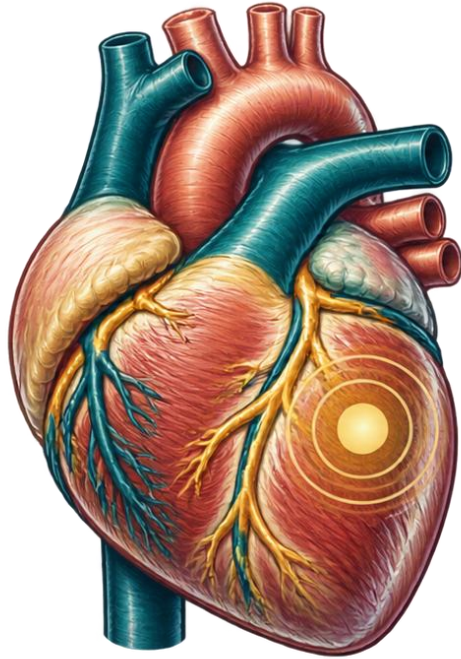
Receptor effects and examples

- β_1 • heart and juxtaglomerular cells: $\uparrow$ HR, contractility and renin; **dobutamine** activates, **metoprolol** blocks.
- β_2 • bronchi, uterus and skeletal muscle vessels: relaxation, bronchodilation and vasodilation; **salbutamol** activates.
- β_3 • adipose tissue and bladder detrusor: lipolysis and detrusor relaxation; **mirabegron** activates.

Clinical link β_2 agonists may cause tremor and can lower serum potassium transiently.

Beta receptor targets in three organs

Match receptor location to the dominant clinical effect



β_1 (heart): rate and contractility β_2 (lungs): bronchodilation β_3 (bladder): relaxation

For storage, α_1 contracts the bladder neck; ***tamsulosin*** relaxes the outlet in BPH.



Organ response: receptor + tissue

A transmitter can produce different effects in different organs

The same transmitter can have opposite effects in different organs because the receptor subtype and tissue differ.

Functional examples

- NE + $\alpha 1$ + arteriole $\rightarrow$ vasoconstriction.
- **Epinephrine** + $\beta 2$ + bronchus $\rightarrow$ relaxation.
- ACh + M2 + heart $\rightarrow$ slowing.

Cardiovascular effects

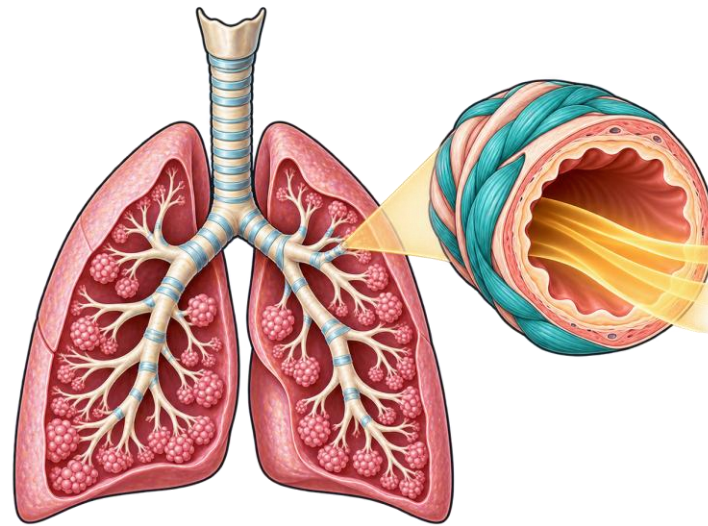
- $\beta 1$ raises heart rate, contractility and conduction.
- $\alpha 1$ increases peripheral resistance through vasoconstriction.
- M2 slows sinus rate and AV conduction.
- Direct parasympathetic control of most peripheral vessels is limited.

Airway receptors at the bronchiole

Drug targets explain bronchodilation and antimuscarinic treatment

β_2 agonist

Salbutamol / albuterol → smooth muscle relaxation → wider airway



M3 antagonist

Ipratropium* or *tiotropium → less vagal bronchoconstriction

Gastrointestinal autonomic effects

Enteric circuits generate local activity under ANS modulation

Enteric circuits generate local motility and secretory patterns. Sympathetic and parasympathetic input modifies their activity.

Gastrointestinal responses

- Parasympathetic activity promotes peristalsis, secretion and digestion.
- Sympathetic activity generally reduces motility, secretion and intestinal blood flow.
- Sympathetic stimulation contracts many gastrointestinal sphincters.
- Myenteric and submucosal reflexes can operate locally and receive SNS/PNS modulation.

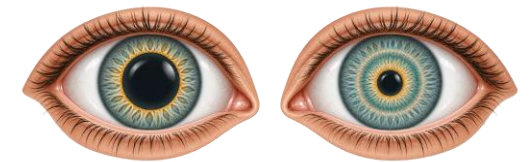
Autonomic control of the eye

Radial and circular iris muscles produce opposing pupil responses

The iris contains radial and circular muscles. Their opposing autonomic receptors control pupil size and near accommodation.

Pupil and accommodation

- α_1 contracts radial iris muscle $\rightarrow$ mydriasis
- M3 contracts sphincter pupillae $\rightarrow$ miosis;
pilocarpine is a muscarinic agonist.
- M3 contracts ciliary muscle $\rightarrow$ accommodation for near vision.
-



Skin, sweat and temperature

Sweating is sympathetic in anatomy but cholinergic at the gland

Thermoregulatory sweating illustrates a sympathetic pathway that uses acetylcholine at its effector gland.

Skin and heat loss

- Cutaneous vasoconstriction helps conserve heat during cold exposure.
- Sympathetic cholinergic ACh stimulates muscarinic sweat-gland receptors.
- ***Atropine*** may reduce sweating and impair heat loss.

Bladder storage and emptying

Different receptors act on the detrusor and outlet

Storage and voiding require coordinated actions at the detrusor and bladder outlet, so different receptors are useful drug targets.

Storage versus voiding

- M3 contracts detrusor during micturition;
- β_3 relaxes detrusor during storage; **mirabegron** can reduce overactive bladder urgency.
- α_1 contracts bladder neck/prostate; **tamsulosin** relaxes the outlet in BPH.
- Antimuscarinic therapy such as **oxybutynin** can reduce detrusor overactivity.

Bladder receptors and drug targets

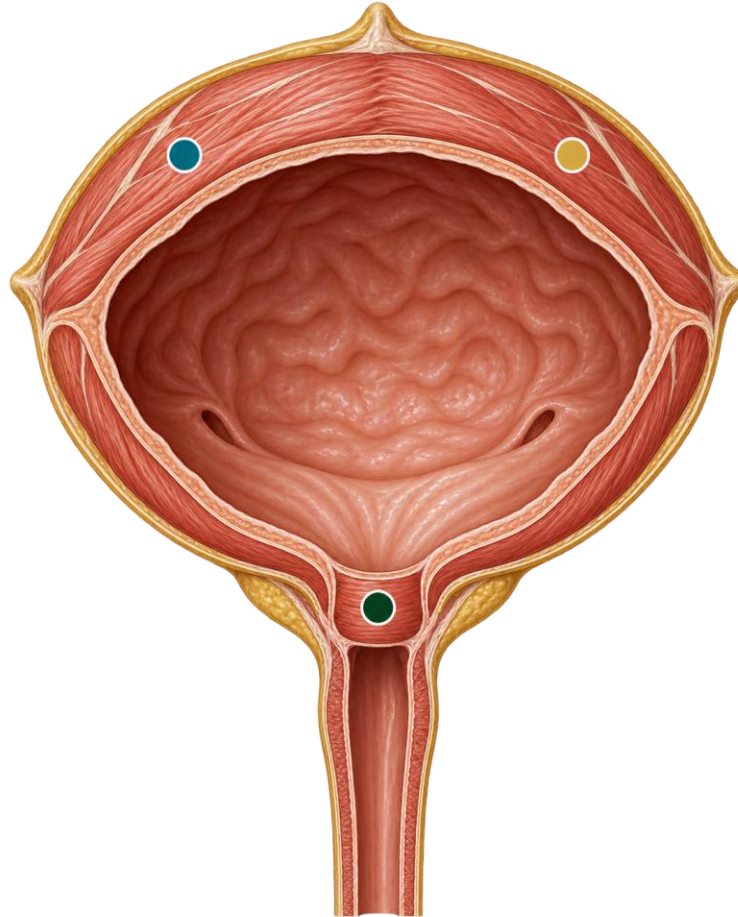
Storage and emptying use the detrusor and outlet differently

β_3 • detrusor

Mirabegron relaxes the bladder wall for storage.

M3 • detrusor

Bethanechol aids contraction;
oxybutynin blocks M3 for urgency.



α_1 • outlet

Tamsulosin blocks α_1 and relaxes the bladder neck/prostate in BPH.

Metabolic effects of sympathetic activation

Catecholamine signals mobilize fuel during acute demand

.

Metabolic responses

- Sympathetic activation increases glycogenolysis and gluconeogenesis.
- Lipolysis releases free fatty acids from adipose tissue.
- Circulating glucose rises to support acute energy use.
- β_2 stimulation also contributes to potassium movement into cells.

Baroreceptor reflex

The brainstem adjusts sympathetic and vagal output when BP changes

The baroreceptor reflex stabilizes arterial pressure by adjusting sympathetic and vagal output in the medulla.

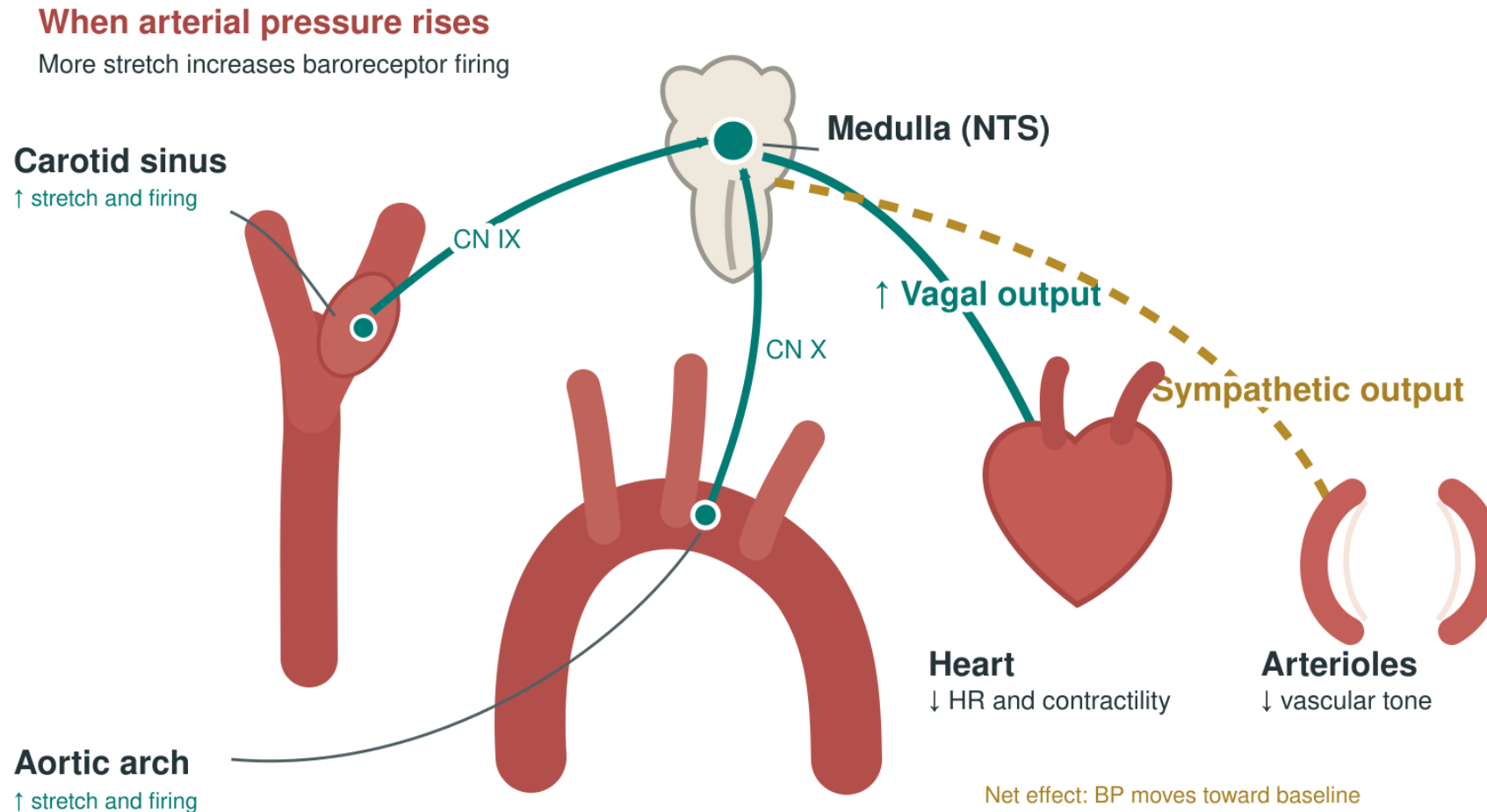
Reflex sequence

- Higher BP stretches carotid sinus and aortic arch baroreceptors.
- CN IX and X afferents signal the medullary NTS.
- Brainstem response: ↓ sympathetic outflow and ↑ vagal outflow.
- Heart rate, contractility and vascular tone fall; impairment may cause labile BP or orthostatic hypotension.

↑ BP → ↑ baroreceptor firing → ↓ SNS + ↑ vagal output → ↓ HR and vascular tone

Baroreceptor reflex: anatomical pathway

The response to rising arterial pressure



Major classes of ANS drugs

Drugs act at receptors or alter transmitter handling

Autonomic drugs act either at the final receptor or at earlier steps that govern transmitter availability.

Major drug classes

- Sympathomimetics activate adrenergic receptors or increase adrenergic transmission.
- Sympatholytics block adrenergic receptors or reduce sympathetic tone.
- Parasympathomimetics activate muscarinic receptors or increase ACh.
- Parasympatholytics block muscarinic receptors.
- Other agents alter synthesis, storage, release, reuptake or metabolism of transmitter.

Sympathomimetics

Receptor selectivity explains both effects and uses

Receptor selectivity determines the dominant organ response and helps match a sympathomimetic drug to its indication.

Representative agonists

- **Epinephrine**: α/β agonist; used in anaphylaxis and cardiac arrest.
- **Norepinephrine**: strong α_1 with β_1 activity; vasopressor in shock.
- **Isoproterenol**: nonselective β agonist; raises HR and bronchodilates.
- **Albuterol/salbutamol** and **salmeterol**: β_2 agonist bronchodilators in asthma/COPD.

Clinical link For anaphylaxis, IM **epinephrine** is the first-line emergency treatment.



Sympatholytics

Blocking α_1 or β receptors produces different clinical effects

Alpha and beta blockade produce different effects on vascular tone, cardiac output, bladder outlet and airway caliber.

Receptor blockade

- α_1 blockers cause vasodilation and relax bladder neck/prostate smooth muscle.
- **Prazosin** is an α_1 blocker; related agents such as **tamsulosin** treat BPH symptoms.
- β blockers lower HR, contractility and renin release; **metoprolol** favors β_1 , while **propranolol** blocks β_1/β_2 .
- Nonselective β blockade can worsen bronchospasm by blocking airway β_2 receptors.



Parasympathomimetics

Direct agonists and AChE inhibitors increase cholinergic effects

Cholinergic effects can be increased directly by a receptor agonist or indirectly by preventing acetylcholine breakdown.

Two approaches

- Direct muscarinic agonists mimic ACh at effector receptors.
- **Bethanechol** stimulates bladder and GI smooth muscle.
- **Pilocarpine** acts on muscarinic receptors; used for xerostomia and in selected glaucoma care.
- Acetylcholinesterase inhibitors raise ACh at cholinergic synapses.

Antimuscarinic drugs

Predict therapeutic actions and adverse effects from M blockade

Muscarinic blockade can be therapeutically useful, but the same receptor actions produce predictable adverse effects.

Actions and adverse effects

- ***Atropine*** blocks M receptors, raises HR by cardiac M₂ blockade and reduces secretions.
- Inhaled ***ipratropium/tiotropium*** block airway muscarinic receptors and dilates bronchi
- Excessive blockade causes antimuscarinic toxicity; reduced sweating may impair heat loss.

Drugs affecting transmitter handling

Autonomic effects can arise before a receptor is reached

Changes in synthesis, storage, reuptake or metabolism alter the amount of ***noradrenaline*** available at the synapse.

Before the receptor

- Catecholamine storage, metabolism and reuptake alter transmitter availability.
- ***Reserpine*** depletes NE stores and is now rarely used.
- MAO inhibitors change catecholamine metabolism by inhibiting MAO enzyme
- Tricyclic antidepressants (TCAs) affect catecholamine reuptake and can cause autonomic adverse effects.



Clinical application: anaphylaxis

Epinephrine addresses airway and circulation through several receptors

Epinephrine addresses several features of anaphylaxis at once because its α and β actions support circulation and airflow.

Why *adrenaline* works

- Pattern: hypotension, bronchospasm, urticaria and possible airway edema.
- Give IM ***epinephrine*** promptly for airway, breathing or circulation compromise.
- α_1 raises vascular tone and reduces mucosal edema; β_1 supports cardiac output.
- β_2 bronchodilates and reduces mediator release; reassess and follow emergency care guidance.

Pathway summary

Link transmitter, receptor and organ to predict the effect

For each pathway, identify the transmitter, receptor and organ before predicting a physiological or drug response.

Four output maps

- Sympathetic: pre ACh $\rightarrow$ Nn; usually post NE $\rightarrow$ α/β .
- Parasympathetic: pre ACh $\rightarrow$ Nn; post ACh $\rightarrow$ M.
- Adrenal medulla: ACh $\rightarrow$ Nn; **epinephrine**/NE into blood.
- Sweat glands: ACh $\rightarrow$ Nn; post ACh $\rightarrow$ M.



Three take-home messages

Remember

- 1** Both divisions use ACh at ganglionic Nn receptors.
- 2** The effector receptor and tissue predict organ response.
- 3** The same map predicts drug benefit and adverse effects.



Thank you